APPLIED MATHEMATICS II

QUADRIC SURFACES

A visual introduction for engineering students



LECTURE NOTES | 3D ANALYTIC GEOMETRY

What is a quadric surface?

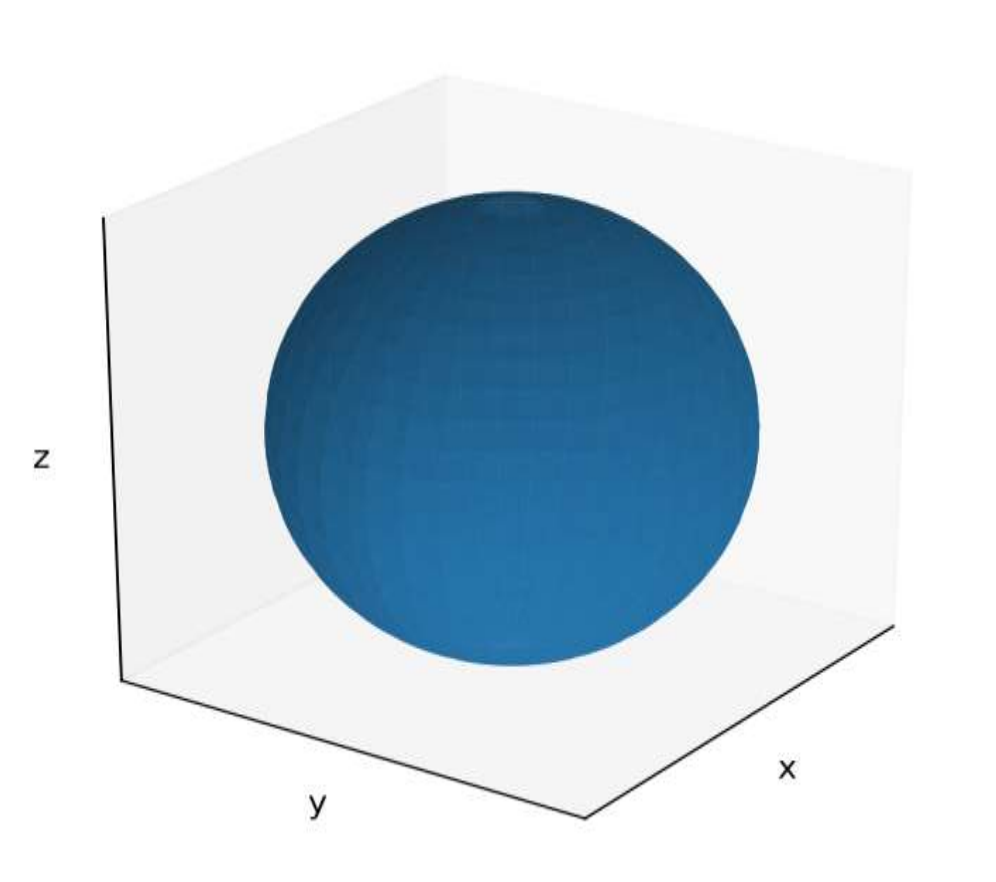
Definition

A quadric surface in 3D is the set of points (x, y, z) satisfying a second-degree equation.

$$Ax^2 + By^2 + Cz^2 + Dxy + Exz + Fyz + Gx + Hy + Iz + J = 0$$

In this lecture, we focus on standard forms with no cross terms (xy, xz, yz).

1. Ellipsoid



Standard form

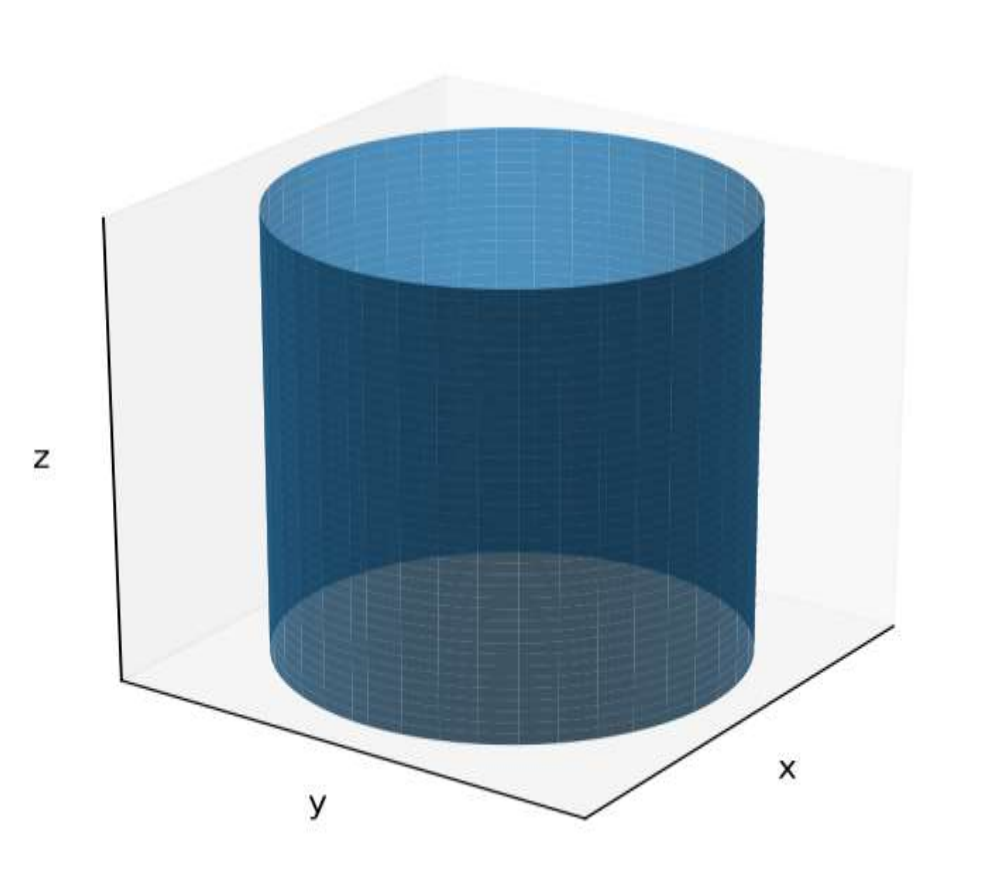
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

- Closed, bounded surface
- Center at the origin
- Semi-axis lengths: a, b, c
- **Sphere** is the special case a = b = c

Example: $\frac{x^2}{9} + \frac{y^2}{4} + z^2 = 1$

Intercepts: (±3,0,0), (0,±2,0), (0,0,±1)

2. Quadric cylinders

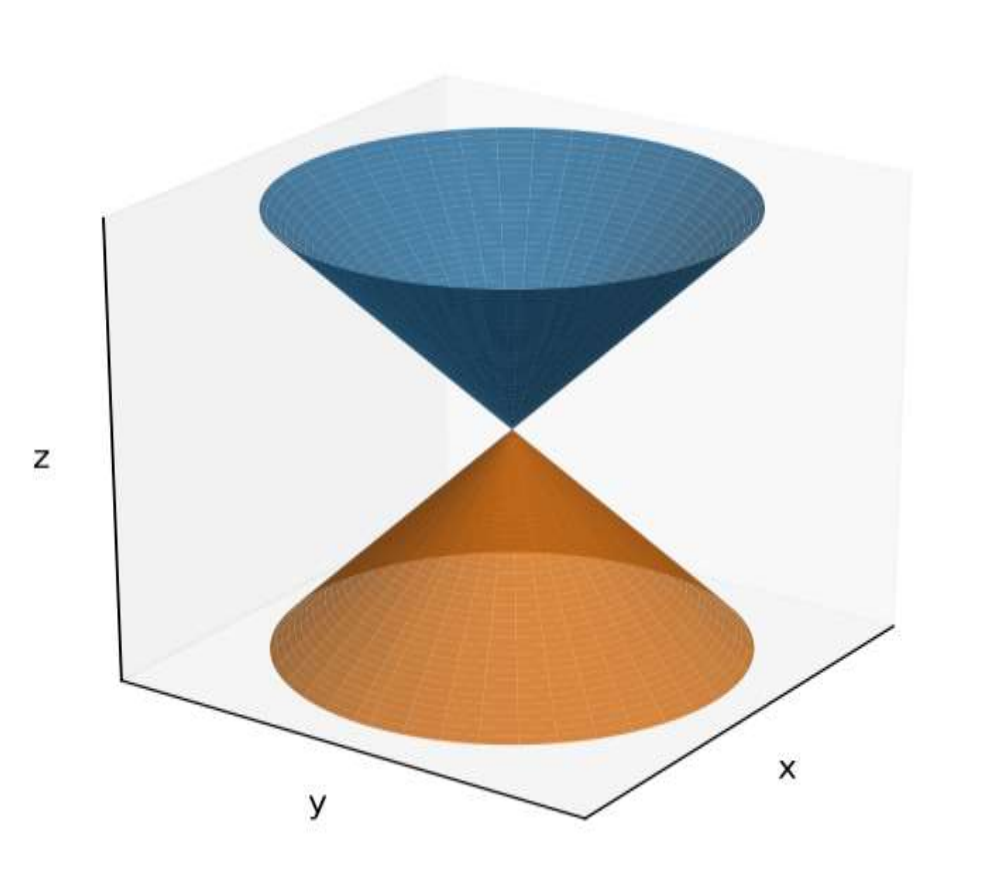


Example

$$x^2 + y^2 = 4$$

- A circular cylinder of radius 2
- z does not appear, so the surface extends parallel to the z-axi
- Replacing 4 by r^2 gives radius r
- $x^2/a^2 + y^2/b^2 = 1$ is an elliptic cylinder

3. Elliptic Double cone



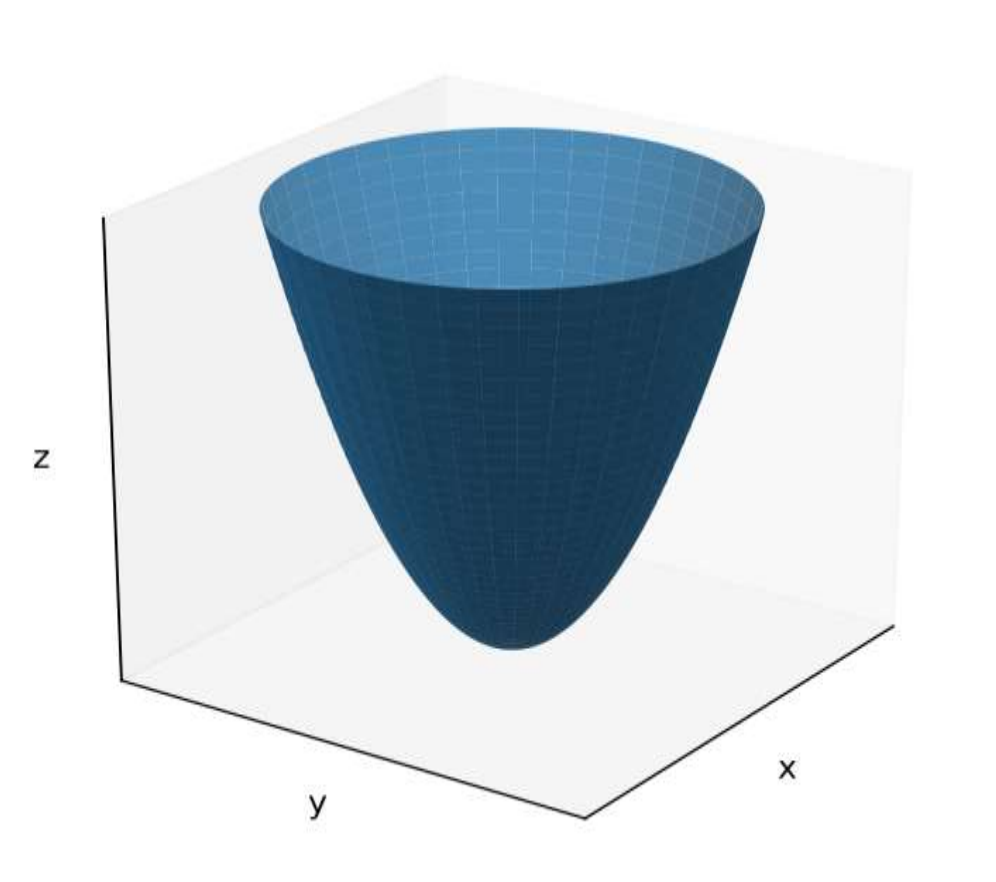
Standard form

$$x^2/a^2 + y^2/b^2 - z^2/c^2 = 0$$

- Double cone meeting at the vertex
- Cross-sections perpendicular to the axis are ellipses
- The equation is homogeneous: all terms are degree 2
- The vertex is at the origin in standard form

Check: $x^2 + y^2 = z^2$ is a circular cone.

4. Elliptic paraboloid



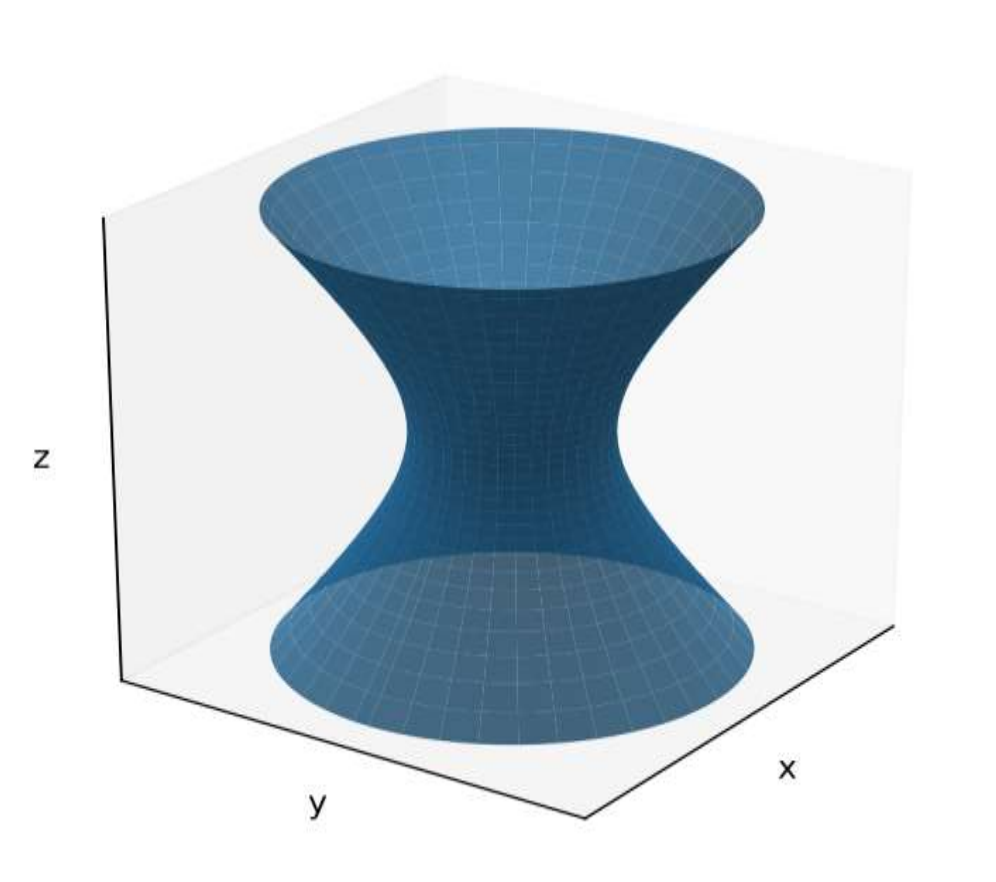
Standard form

$$z = x^2/a^2 + y^2/b^2$$

- Bowl-shaped surface
- Opens in the positive z-direction (as written)
- Horizontal traces $z = k > 0$ are ellipses
- Vertical traces are parabolas

If z is replaced by −z, the bowl opens downward.

5. Hyperboloid of one sheet



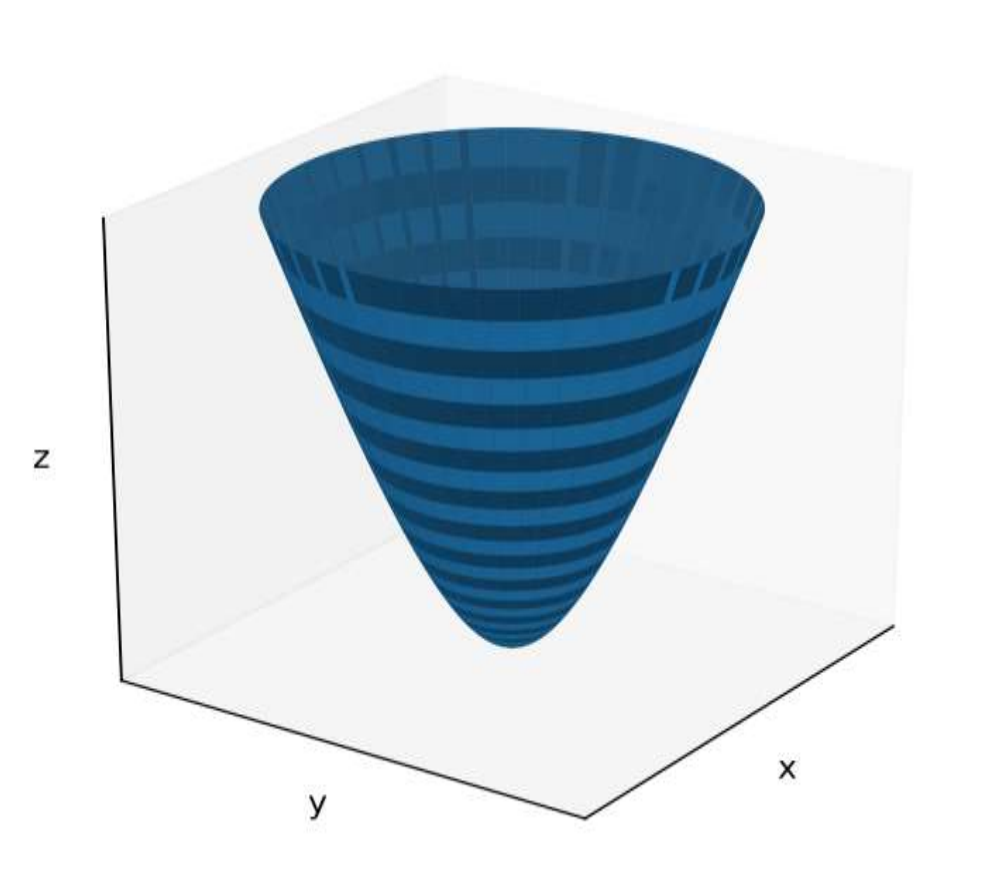
Standard form

$$x^2/a^2 + y^2/b^2 - z^2/c^2 = 1$$

- One connected surface
- The variable with the negative term gives the axis of the surface
- Horizontal traces z = k are ellipses
- Narrowest waist occurs at z = 0

Engineering note: similar geometry appears in cooling towers and som

3. Hyperboloid of two sheets



Standard form

$$z^2/c^2 - x^2/a^2 - y^2/b^2 = 1$$

- Two disconnected pieces (sheets)
- Opens along the axis of the positive squared term
- Exists only where $|z| \geq c$
- At $z = 0$, there are no real points

Tip: one positive term and two negative terms → two sheets when the equation equals 1.

Worked example 1 • classify an ellipsoid

Classify and describe: $4x^2 + y^2 + 9z^2 = 36$

Step 1 Divide both sides by 36.

$$\frac{x^2}{9} + \frac{y^2}{36} + \frac{z^2}{4} = 1$$

Step 2 Match the ellipsoid form.

$a = 3,\ b = 6,\ c = 2$

Conclusion

An ellipsoid centered at (0,0,0), with semi-axes 3 along x, 6 along y, and 2 along z.

Intercepts: $x = \pm 3$, $y = \pm 6$, $z = \pm 2$ (set the other two variables to zero).

Worked example 2 • complete the square

Rewrite and classify: $x^2 + y^2 + z^2 - 6x + 4y - 2z = 11$

Group variables and move the constant:

$(x^2 - 6x) + (y^2 + 4y) + (z^2 - 2z) = 11$

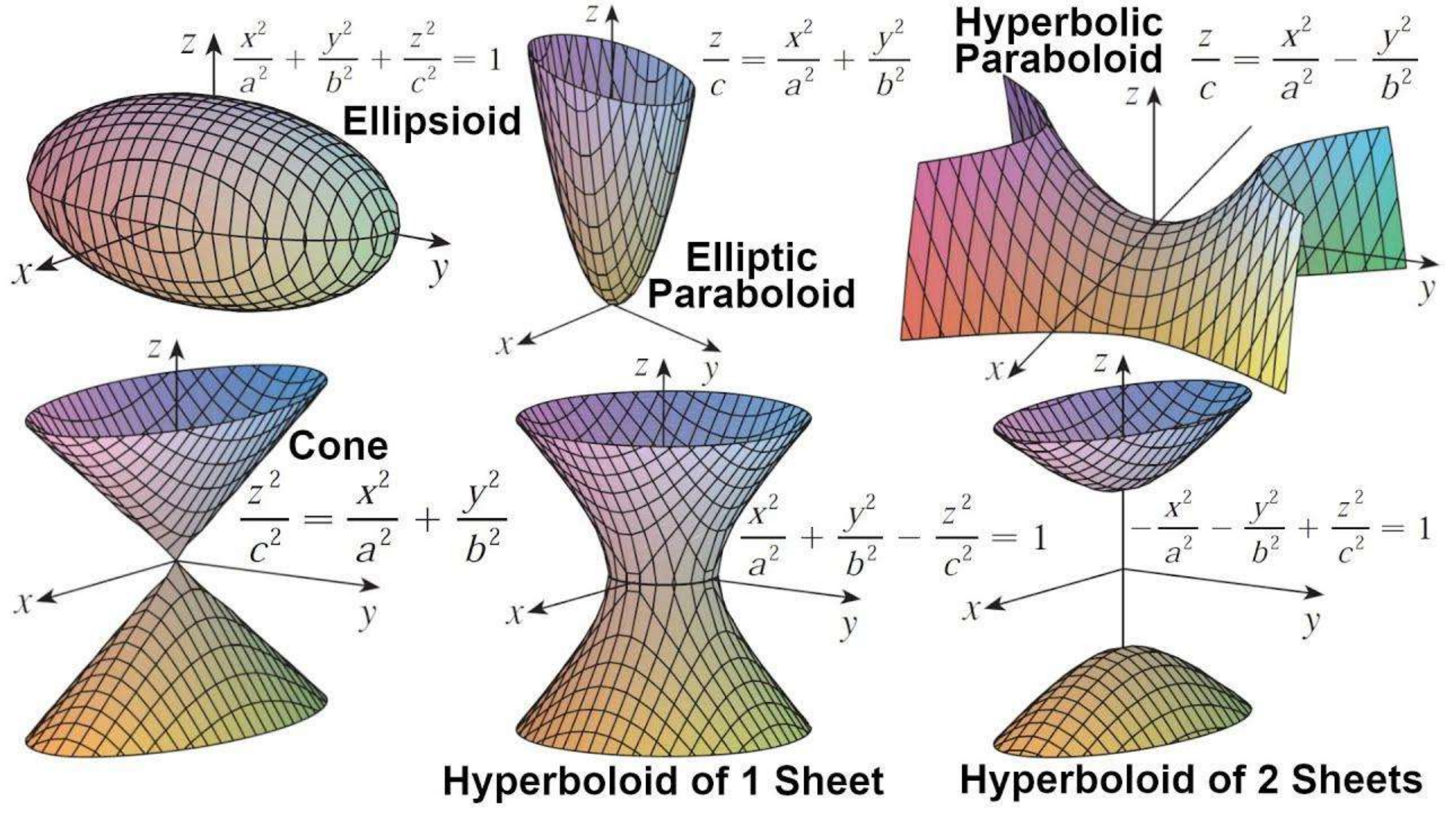
Complete each square:

$(x - 3)^2 - 9 + (y + 2)^2 - 4 + (z - 1)^2 - 1 = 11$

$$\mathbf{(x - 3)^2 + (y + 2)^2 + (z - 1)^2 = 25}$$

Conclusion

Sphere of radius 5 centered at (3, −2, 1).


$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$
Ellipsoid
$\frac{z}{c} = \frac{x^2}{a^2} + \frac{y^2}{b^2}$
Elliptic Paraboloid
Hyperbolic Paraboloid
$\frac{z}{c} = \frac{x^2}{a^2} - \frac{y^2}{b^2}$
Cone
$\frac{z^2}{c^2} = \frac{x^2}{a^2} + \frac{y^2}{b^2}$
$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$
Hyperboloid of 1 Sheet
$-\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$
Hyperboloid of 2 Sheets

Quick classification guide

Surface	Signature / standard form	Shape clue
Ellipsoid	$x^2/a^2 + y^2/b^2 + z^2/c^2 = 1$	Closed
One-sheet hyperboloid	+ + − = 1	One connected piece
Two-sheet hyperboloid	+ − − = 1 (one positive term)	Two pieces
Elliptic cone	+ + − = 0	Double cone
Elliptic paraboloid	$z = x^2/a^2 + y^2/b^2$	Bowl
Cylinder	one variable absent	Extends along an axis

Try it • guided practice

A $x^2/16 + y^2/9 + z^2/4 = 1$

Name the surface and its semi-axes.

B $x^2 + y^2 - z^2 = 0$

Name the surface.

C $z = 9 - x^2 - y^2$

Describe the opening direction and vertex.

D $x^2 + y^2 = 25$

Name the surface and its axis.